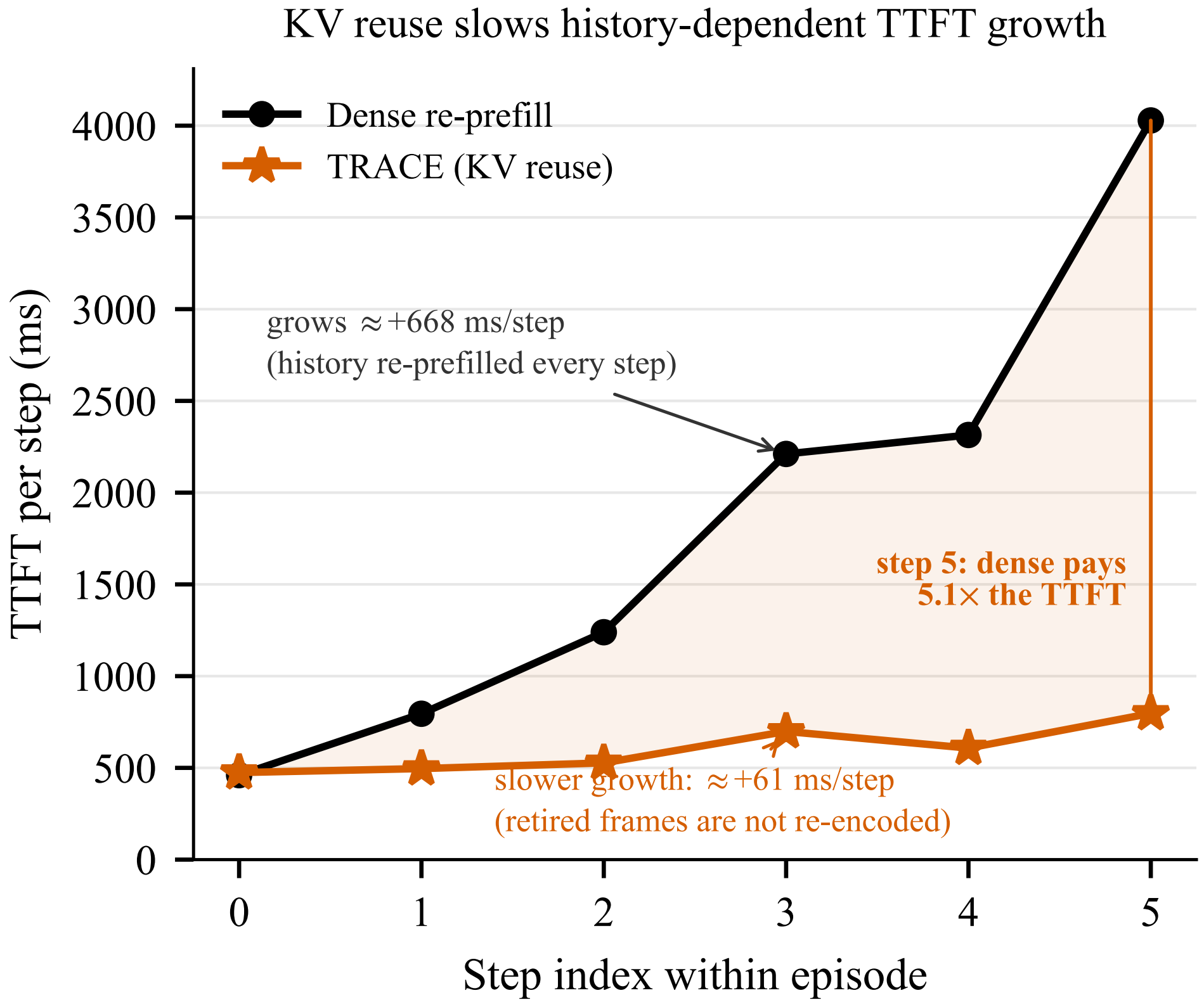

KV reuse slows history-dependent TTFT growth
Dense re-prefill
TRACE (KV reuse)
TTFT per step (ms)
4000
3500
3000
2500
2000
1500
1000
500
0
grows ≈+668 ms/step
(history re-prefilled every step)
step 5: dense pays
5.1× the TTFT
slower growth: ≈+61 ms/step
(retired frames are not re-encoded)
0
1
2
3
4
5
Step index within episode